Supplementary Material for UNT-STLLM

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I. Reproduction Details

This supplement gives resource measurement and adaptation details, paired error comparisons, complete transfer metrics, and attention diagnostics. Accompanying Source Data files contain the individual runs.

A. Resource Measurement

Inference used the model state selected by validation MAE after optimizer and training states were released. Batch size was 20, with three warmup batches and ten timed batches per run. CUDA synchronization bracketed each timed batch. Each run contributed its median batch latency, and throughput was batch size divided by this median. Training peaks refer to allocated memory during training passes. Time per epoch included optimizer initialization in the first epoch. Complete unit time also included evaluation before result serialization, as specified in the main resource figure.

B. Target Adaptation

Each source stage used its complete training split. Both target initializations received the same fixed 260 training windows within seed, with an identical scaler fitted to those target inputs. Target node embeddings and static relation matrices were initialized afresh. Source initialization reused all other learned parameters, while independent initialization used fresh forecasting modules with pretrained GPT-2 weights. Target stages shared random states and batch order within seed, starting with empty optimizer states. Each stage ran for 250 epochs with batch 20. Each target stage performed 3,250 optimizer updates, with validation MAE selecting its model state. Source test data were excluded from source model selection.

C. Representation Summaries

The main attention, gate, and PCA analyses used the earlier evaluation configuration selected by validation MAE. Seeds were 2028 for taxi pick-up, 2027 for taxi drop-off, 2029 for bicycle pick-up, and 2027 for bicycle drop-off. Attention and gate summaries used all 870 validation windows, with attention averaged across heads. Gate weights were grouped by the final observed hour. PCA used all nodes from eight evenly spaced validation windows and was fitted separately for each task.

II. Paired Error Comparisons

Comparisons are paired within seed. Forecasting gains subtract the full model error from the reference error. Adaptation gains subtract transferred initialization error from independent initialization error. The 95% intervals in Table S1 are $\overline{d} \pm t_{0.975,4} s_d/\sqrt{5}$, where s_d is the sample standard deviation of five paired differences. These intervals summarize training variation across seeds. Relative gains divide differences in aggregate means by reference means. WAPE differences are percentage points. The Source Data retain both ratios of means and means of paired percentage gains.

TABLE S1. Paired improvements in forecasting error. Positive differences indicate lower error for UNT-STLLM relative to the listed reference, or for transferred relative to independent initialization. Intervals summarize seed variation over five paired runs. MAE/RMSE use flow counts, whereas WAPE differences use percentage points.

Comparison	Task or direction	Metric	Mean gain	95% interval	Gain (%)
Base	Taxi drop-off	MAE	0.1093	[0.0311, 0.1876]	2.18
Base	Taxi drop-off	RMSE	0.1793	[0.0281, 0.3305]	2.04
Base	Taxi drop-off	WAPE	0.4172	[0.1186, 0.7159]	2.18
ST-LLM+	Taxi drop-off	MAE	0.1136	[-0.0235, 0.2507]	2.26
ST-LLM+	Taxi drop-off	RMSE	0.2040	[0.0501, 0.3580]	2.32
ST-LLM+	Taxi drop-off	WAPE	0.4333	[-0.0897, 0.9563]	2.26
Base	Bicycle drop-off	MAE	0.0014	[-0.0057, 0.0085]	0.07
Base	Bicycle drop-off	RMSE	-0.0072	[-0.0268, 0.0123]	-0.26
Base	Bicycle drop-off	WAPE	0.0274	[-0.1163, 0.1711]	0.07
ST-LLM+	Bicycle drop-off	MAE	0.0232	[0.0071, 0.0392]	1.22
ST-LLM+	Bicycle drop-off	RMSE	0.0669	[0.0261, 0.1077]	2.35
ST-LLM+	Bicycle drop-off	WAPE	0.4682	[0.1441, 0.7923]	1.22
Base	Taxi pick-up	MAE	0.0968	[0.0338, 0.1597]	1.86

Comparison	Task or direction	Metric	Mean gain	95% interval	Gain (%)
Base	Taxi pick-up	RMSE	0.1353	[-0.0166, 0.2872]	1.51
Base	Taxi pick-up	WAPE	0.3661	[0.1281, 0.6040]	1.86
ST-LLM+	Taxi pick-up	MAE	0.0861	[0.0407, 0.1314]	1.66
ST-LLM+	Taxi pick-up	RMSE	0.2091	[0.1066, 0.3116]	2.31
ST-LLM+	Taxi pick-up	WAPE	0.3255	[0.1541, 0.4969]	1.66
Base	Bicycle pick-up	MAE	-0.0088	[-0.0186, 0.0011]	-0.44
Base	Bicycle pick-up	RMSE	-0.0256	[-0.0492, -0.0021]	-0.84
Base	Bicycle pick-up	WAPE	-0.1766	[-0.3756, 0.0223]	-0.44
ST-LLM+	Bicycle pick-up	MAE	0.0099	[-0.0068, 0.0265]	0.49
ST-LLM+	Bicycle pick-up	RMSE	0.0173	[-0.0018, 0.0364]	0.56
ST-LLM+	Bicycle pick-up	WAPE	0.1984	[-0.1367, 0.5335]	0.49
Transferred	Taxi pick-up→drop-off	MAE	0.4576	[0.3857, 0.5295]	8.12
Transferred	Taxi pick-up→drop-off	RMSE	0.9484	[0.7670, 1.1298]	9.40
Transferred	Taxi pick-up→drop-off	WAPE	1.7462	[1.4721, 2.0203]	8.12
Transferred	Taxi drop-off→pick-up	MAE	0.4343	[0.3529, 0.5158]	7.42
Transferred	Taxi drop-off→pick-up	RMSE	0.7695	[0.3193, 1.2197]	7.49
Transferred	Taxi drop-off→pick-up	WAPE	1.6432	[1.3351, 1.9512]	7.42
Transferred	Bicycle pick-up→drop-off	MAE	0.0889	[0.0791, 0.0988]	4.42
Transferred	Bicycle pick-up→drop-off	RMSE	0.1954	[0.1721, 0.2187]	6.39
Transferred	Bicycle pick-up→drop-off	WAPE	1.7981	[1.5995, 1.9968]	4.42
Transferred	Bicycle drop-off→pick-up	MAE	0.0930	[0.0756, 0.1104]	4.37
Transferred	Bicycle drop-off→pick-up	RMSE	0.1996	[0.1674, 0.2318]	5.94
Transferred	Bicycle drop-off→pick-up	WAPE	1.8731	[1.5230, 2.2232]	4.37

III. Environment Consistency

The two environments used PyTorch 2.8.0 with CUDA 12.8, with hardware assignments given in the main setup. Table S2 compares common configurations across these environments. Each controlled comparison retains its own environment reference.

TABLE S2
MAE means with sample standard deviations over five seeds for both execution environments. Percentage changes compare the environment B mean with the environment A mean. Hardware assignments appear in the main experimental setup.

Condition	Task	Environment A	Environment B	Change (%)
Full model	Taxi drop-off	4.9077 ± 0.0380	4.9279 ± 0.0592	+0.411
Full model	Bicycle drop-off	1.8793 ± 0.0036	1.8828 ± 0.0094	+0.187
Zero output, rank 8	Taxi drop-off	4.9326 ± 0.0457	4.9557 ± 0.0701	+0.469
Zero output, rank 8	Bicycle drop-off	1.8816 ± 0.0080	1.8809 ± 0.0082	-0.039
Zero output, rank 64/62	Taxi drop-off	4.9624 ± 0.0211	4.9661 ± 0.0507	+0.074
Zero output, rank 64/62	Bicycle drop-off	1.8833 ± 0.0058	1.8822 ± 0.0090	-0.056

IV. Complete Adaptation Metrics

Table S3 supplies the three forecasting metrics for the four directions in the main transfer figure. All target comparisons share the fixed support and optimization budget described above.

TABLE S3
Adaptation in four directions under fixed 10% target support

Direction	Initialization	MAE	RMSE	WAPE (%)
Taxi pick-up → drop-off	Independent	$\underline{5.633} \pm 0.044$	$\underline{10.093} \pm 0.058$	$\underline{21.50} \pm 0.17$
Taxi pick-up → drop-off	Transferred	$\mathbf{5.176} \pm 0.067$	$\mathbf{9.145} \pm 0.108$	$\mathbf{19.75} \pm 0.26$
Taxi drop-off → pick-up	Independent	$\underline{5.850} \pm 0.086$	$\underline{10.270} \pm 0.349$	$\underline{22.13} \pm 0.33$
Taxi drop-off → pick-up	Transferred	$\mathbf{5.416} \pm 0.042$	$\mathbf{9.500} \pm 0.162$	$\mathbf{20.49} \pm 0.16$
Bicycle pick-up → drop-off	Independent	$\underline{2.013} \pm 0.005$	$\underline{3.057} \pm 0.014$	$\underline{40.69} \pm 0.10$
Bicycle pick-up → drop-off	Transferred	$\mathbf{1.924} \pm 0.010$	$\mathbf{2.862} \pm 0.023$	$\mathbf{38.89} \pm 0.20$
Bicycle drop-off → pick-up	Independent	$\underline{2.127} \pm 0.012$	$\underline{3.360} \pm 0.021$	$\underline{42.84} \pm 0.24$
Bicycle drop-off → pick-up	Transferred	$\mathbf{2.034} \pm 0.005$	$\mathbf{3.160} \pm 0.011$	$\mathbf{40.96} \pm 0.10$

Entries are mean ± sample standard deviation over five seeds with the same fixed 260 target windows within seed, with lower values preferred, bold indicating the best and underlining the second best within each direction.

V. Attention Diagnostics

Table S4 summarizes validation weights after softmax for the two post-softmax conditions. Each seed combines the final two relation layers using element counts for negative fractions, row counts for row errors, and masked entry counts for masked magnitudes. Means and sample standard deviations are then computed over five seeds. Diagnostics include the padded final validation batch. Bidirectional conditions have no masked entries. The accompanying CSV files retain all 40 layer records and 20 seed summaries.

TABLE S4
VALIDATION DIAGNOSTICS FOR RELATION WEIGHTS ADDED AFTER SOFTMAX. ENTRIES ARE MEANS WITH SAMPLE STANDARD DEVIATIONS OVER FIVE SEEDS. NEGATIVE WEIGHTS ARE PERCENTAGES, ROW ERRORS ARE ABSOLUTE DEVIATIONS FROM UNIT SUMS, AND MASKED WEIGHTS ARE MEAN ABSOLUTE VALUES. DASHES DENOTE THE ABSENCE OF MASKED ENTRIES.

Access	Task	Negative (%)	Row error	Masked weight
Causal	Taxi	69.582 ± 6.736	7.243 ± 1.513	0.032 ± 0.006
Causal	Bicycle	48.697 ± 6.804	8.585 ± 3.127	0.047 ± 0.011
Bidirectional	Taxi	69.094 ± 3.756	6.260 ± 0.708	—
Bidirectional	Bicycle	45.956 ± 7.561	5.113 ± 0.453	—

Fig. S1 extends the main taxi pick-up relation maps to taxi drop-off under the same comparison procedure.

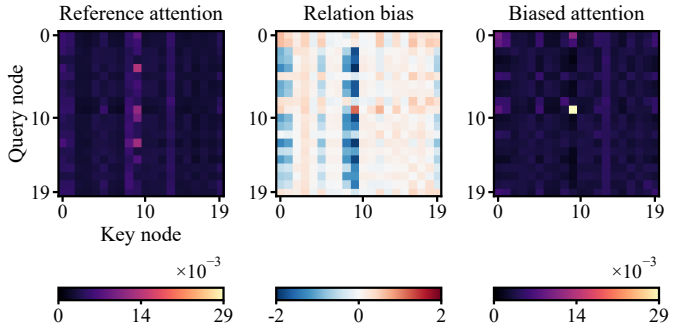


Fig. S1. Taxi drop-off attention in block 5 before and after relation bias injection at a fixed block input. Nodes 0–19 are displayed from 266 nodes, with attention normalized over all nodes. Bias rows are centered for display. Outer color bars give attention weights in units of 10^{-3}, while the middle bar gives bias values.